Meat and fish consumption and cancer in Canada.

In this study, we examined the association between meat and fish intake and the risk of various cancers. Mailed questionnaires were completed by 19,732 incident, histologically confirmed cases of cancer of the stomach, colon, rectum, pancreas, lung, breast, ovary, prostate, testis, kidney, bladder, brain, non-Hodgkin's lymphomas (NHL), and leukemia and 5,039 population controls between 1994 and 1997 in 8 Canadian provinces. Measurement included information on socioeconomic status, lifestyle habits, and diet. A 69-item food frequency questionnaire provided data on eating habits 2 yr before data collection. Odds ratios and 95% confidence intervals were derived through unconditional logistic regression. Total meat and processed meat were directly related to the risk of stomach, colon, rectum, pancreas, lung, breast (mainly postmenopausal), prostate, testis, kidney, bladder, and leukemia. Red meat was significantly associated with colon, lung (mainly in men), and bladder cancer. No relation was observed for cancer of the ovary, brain, and NHL. No consistent excess risk emerged for fish and poultry, which were inversely related to the risk of a number of cancer sites. These findings add further evidence that meat, specifically red and processed meat, plays an unfavorable role in the risk of several cancers. Fish and poultry appear to be favorable diet indicators.